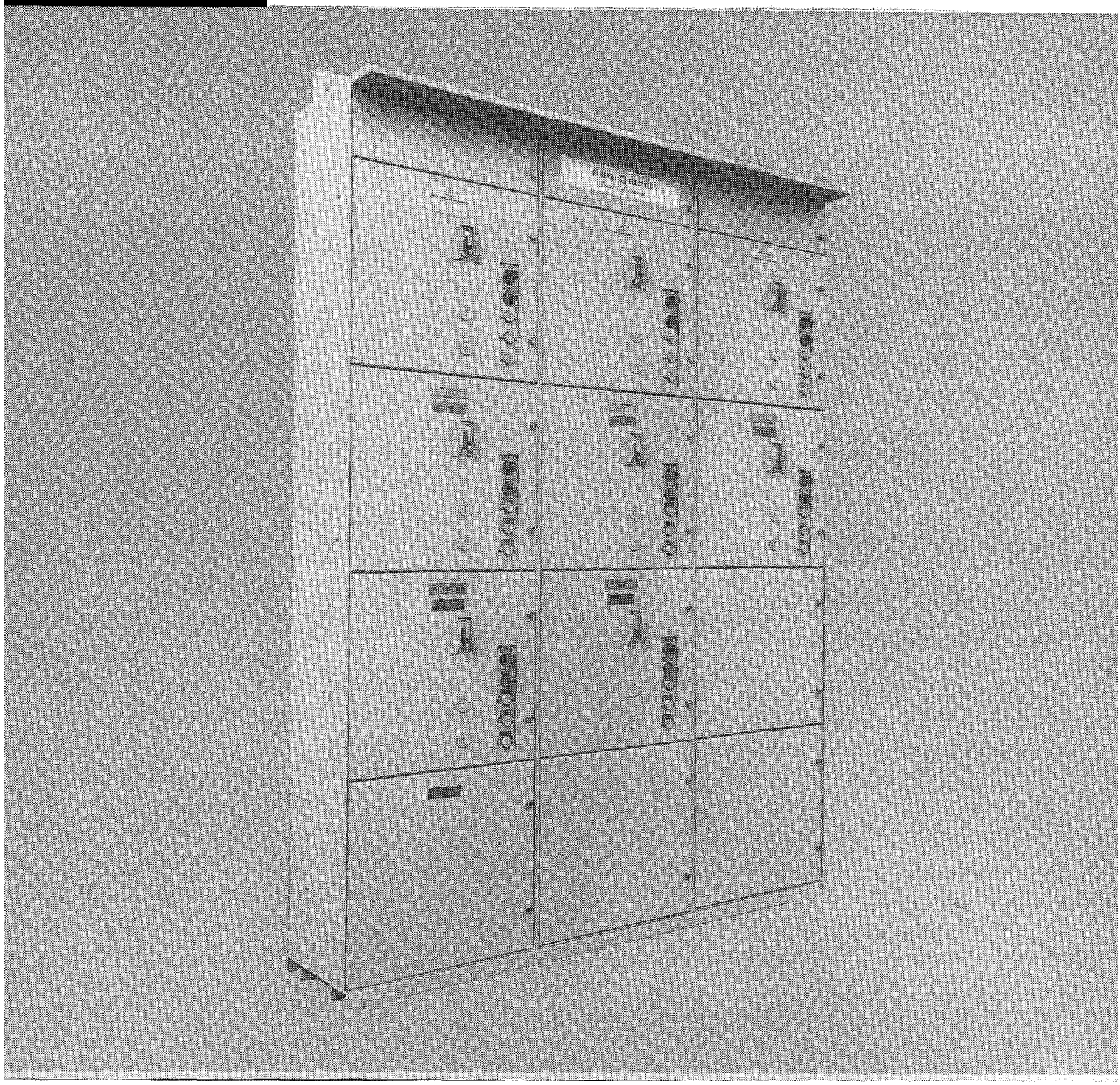
INSTRUCTIONS

GEK-40590

NAVY AC CONTROL CENTERS

TYPE IC5142



GENERAL ELECTRIC

CONTENTS

		Page
SECTION 1, GENERAL INFORMATION		3
1.1	General Data	3
1.2	Introduction and Scope	4
1.3	Warranty Information	4
1.4	Equipment Configuration	4
SECTION 2, OPERATION		4
2.1	Introduction	4
2.2	Controller Unit Operation	4
2.3	Circuit Breaker Operation (Includes Distribution Units)	5
SECTION 3, FUNCTIONAL DESCRIPTION		6
3.1	Enclosure	6
3.2	Buswork	6
3.3	Controller Units	6
3.4	Distribution Units	7
3.5	Equipment Cabling	7
3.6	Auxiliary Relays	8
3.7	Drawout Features	8
SECTION 4, SCHEDULED MAINTENANCE		8
4.1	Controller Units	8
4.2	Circuit Breakers	9
4.3	Controller Unit Removal	9
4.4	Controller Unit Replacement	9
SECTION 5, TROUBLESHOOTING		9
5.1	General	9
5.2	Troubleshooting Chart	9
SECTION 6, CORRECTIVE MAINTENANCE		10
6.1	Introduction	10
6.2	Controller Unit	10
6.3	Individual Device Repair	10
6.4	Circuit Breaker Maintenance	10
SECTION 7, PARTS LIST		10
7.1	General	10
SECTION 8, INSTALLATION		11
8.1	General	11
8.2	Shipping and Handling	11
8.3	Mounting	11
8.4	Incoming Cabling	11
8.5	Door and Cover Diagrams	12
8.6	Preparation for Initial Operation	12

NAVY AC CONTROL CENTERS
TYPE IC5142

SECTION I
GENERAL INFORMATION

1.1 GENERAL DATA

The following reference data, Table 1-1, summarizes the pertinent technical and design characteristics of this equipment:

TABLE 1-1
NAVY AC CONTROL CENTERS

GENERAL	
General Electric Type No.	IC5142B.
Military Specification	Designed and tested to MIL-C-23742.
Power Requirements	440 volts ac, 3 phase, 60 hertz.
Quiet Operation	Designed and tested to MIL-STD-740 when specified.
Ambient Temperature	50 degrees Centigrade.
ENCLOSURE	
Type	Dripproof, inclination up to 45 degrees.
Dimensions	Each vertical section: 79-inch high, 20-inch wide, 12-inch deep.
Unit Compartments	Separate compartments. Compartments separated by steel barriers.
Incoming Power	Provisions in bottom (main bus) compartment.
Load and Control Connections	Bottom cable access. 1. Type B Wiring – Vertical trough provides access to terminal blocks located opposite each individual unit. 2. Type C Wiring – Master terminal boards located in bottom (main bus) compartment.
Access	Front access only. Separate doors for each controller unit and for main bus compartment. Removable covers for each distribution unit.
CONTROLLER UNITS	
Design	Full draw-out construction. Units of the same size and design are readily interchangeable. Standard Navy devices and ratings in accordance with MIL-C-2212.
Overload Protection	Externally reset, ambient compensated, overload relays.
Short Circuit Protection	Line fuses (Size 0 and 1 only) or Type AQB Circuit Breakers (fused or unfused).
Type Motor Load	Squirrel cage, induction – 440 volts ac, 3 phase, 60 hertz.
Fuseholders	Indicating type; fuse is replaceable through door cutout.
Circuit Breaker Operation	Operation through door cutout with guard to minimize accidental operation.
DISTRIBUTION UNITS	
Circuit Breakers	Fused or unfused Type AQB per MIL-C-17361.
Design	Removable mounting circuit breakers. Operation through cutout in removable front panel. Guards to minimize accidental operation.

1.2 INTRODUCTION AND SCOPE

These control centers are designed for Naval Shipboard application and meet the requirements of MIL-C-23742 specifications. They are made up of standardized vertical sections housing the vertical and horizontal bus, vertical wiring troughs, and the compartmented control units. Sections are bolted together to form a single line-up assembly and may contain all controller units, all distribution units, or combinations of both. The controller units supply and control power to the electrical auxiliaries; the distribution units, containing circuit breakers, distribute power to miscellaneous ship service loads.

These instructions pertain to the operation, maintenance, troubleshooting and repair of this equipment and, when supplemented with the following, provide the necessary field data normally required to sustain continued satisfactory operation:

1. Equipment drawings.
2. General Electric instruction publication GEK-40589, "Navy AC and DC Magnetic Starters."
3. General Electric device instruction publications.
4. General Electric publication GEK-40591, "Parts List for AC and DC Magnetic Starters."
5. Appropriate manufacturer's circuit breaker instructions.

These instructions do not purport to cover all details or variations in equipment, nor to provide every possible contingency to be met in connection with installation, operation, or maintenance. Should further information be desired, or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to the General Electric Company, Industrial Control Products Department, Salem, Virginia 24153 (Federal Supply Manufacturer's Code 02989).

NOTE: *When contacting the factory, always mention the General Electric catalog number and drawing number, in addition to the IC Type number, to enable the factory to identify the exact starter configuration in question.*

1.3 WARRANTY INFORMATION

Refer to the particular contract for warranty in effect for equipment.

1.4 EQUIPMENT CONFIGURATION

Equipment configuration is completely covered by the GE drawings furnished with the equipment.

SECTION 2
OPERATION

2.1 INTRODUCTION

Figure 2-1 illustrates a typical control center as seen by the operator. All operator's controls and indicators are accessible from the front. Selected pilot devices have tamperproof covers, when specified, to prevent unauthorized operation.

2.2 CONTROLLER UNIT OPERATION

The following is basically the procedure used to energize and de-energize a controller unit. Refer to the detailed equipment drawings and appropriate system technical manual, if available, for specific description of operation.

1. Energization

WARNING: *ENSURE THAT THE CONTROLLER UNIT IS FULLY INSERTED BEFORE ENERGIZING. SEE PARAGRAPH 4.4 FOR CONTROLLER UNIT REPLACEMENT PROCEDURES.*

a. Ensure that power is available at the control center incoming bus.

b. Turn circuit breaker to the ON position. (Omit this step if controller unit has line fuses in lieu of circuit breaker.)

c. Initiate the local or remote master switch.

d. The controller unit will now be ON unless there is, or has been, a motor-overload condition which would cause the overload relay to trip, opening the control circuit.

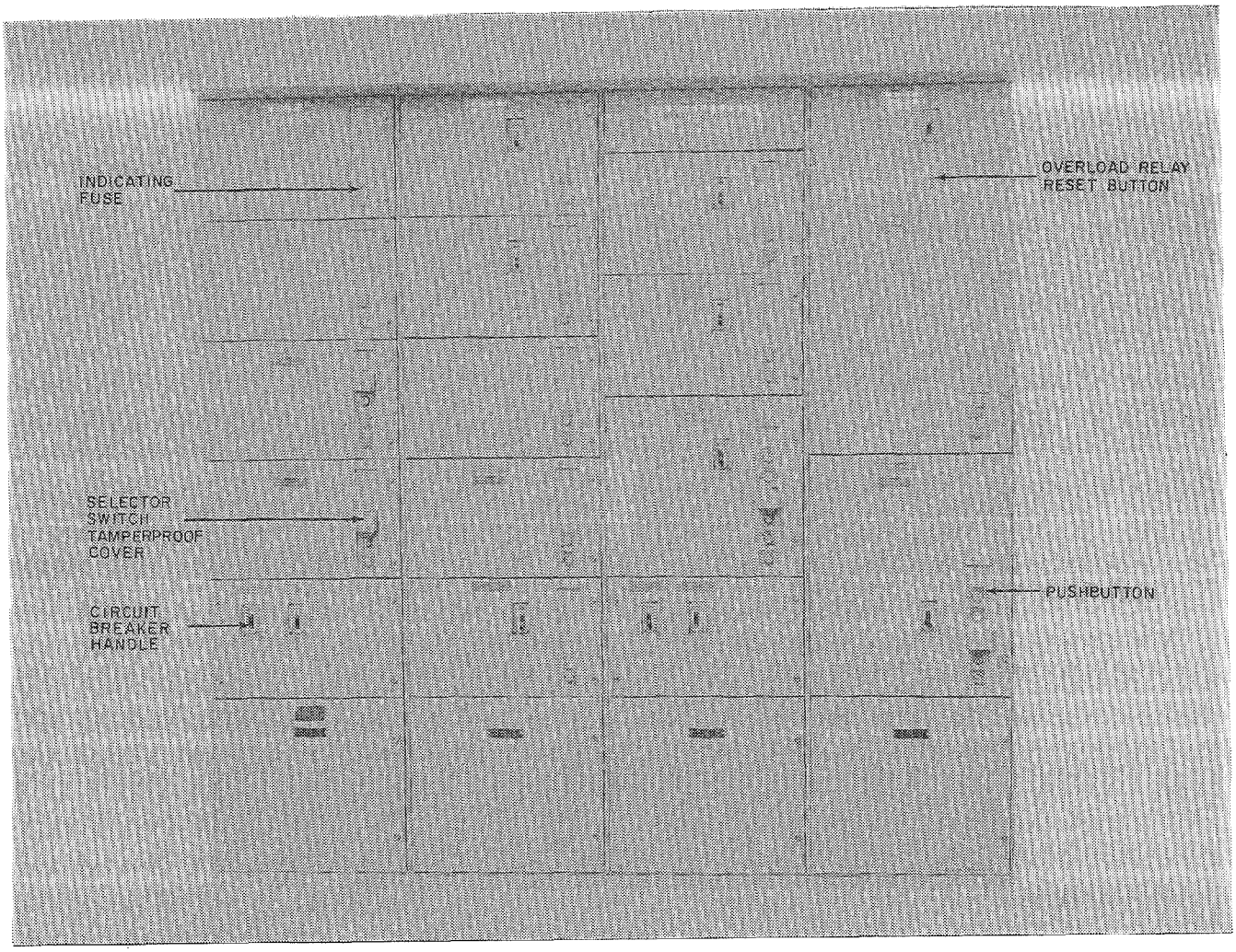


Fig. 2-1. Operator's view of typical control center

e. If the motor does not start after performing step d, push the overload reset button on the door and then repeat step c.

2. De-energization

a. Initiate the local or remote master switch.

b. Turn circuit breaker to the OFF position.

2.2.1 Individual Device Operation

The operation of principal individual devices is contained in the appropriate device instruction publications.

2.3 CIRCUIT BREAKER OPERATION (INCLUDES DISTRIBUTION UNITS)

For information on the operation of circuit breakers, refer to the manufacturer's instruction publications.

SECTION 3
FUNCTIONAL DESCRIPTION

3.1 ENCLOSURE

The standard IC5142 control center vertical section is 79-inch high, 20-inch wide, and 12-inch deep (outside dimensions). Sections are fabricated from sheet steel, shaped, and reinforced to form a rigid, self supporting, dripproof structure in single or multiple section lineups. Single vertical sections are bolted together to form a multiple-section lineup.

Internally, each control center section is subdivided into compartments into which completely drawout controller units, or circuit breakers, are inserted. Standard compartment sizes are 12-inch, 18-inch, and 24-inch high. Modular design, based on 6-inch spacing increments, enables a vertical section to be subdivided, upon assembly, into any desired combination of compartment sizes not exceeding 60-inch in total combined height.

Each vertical section has a full-height vertical wiring area located along the right side of the enclosure to permit wiring access to individual units. The barrier (or shelf), which subdivides the vertical section into compartments, contains knockouts for "feed-through" wiring. In addition, the narrow space (approximately 0.5-inch wide by 8-inch deep) between the shelf and right-hand side of the enclosure provides an area in which smaller diameter wires may be fed through. See paragraph 8.4 for details on installation of wiring and cabling.

3.2 BUSWORK

The 600-amperes vertical buswork (sandwiched between molded, red, glass polyester insulation) is located in the rear of the enclosure and connects to the main buswork in the main bus compartment (see Fig. 8-2). The main buswork is the interconnection buswork between the vertical sections. The main buswork is rated as required to carry the total connected load. Incoming power cables connect to the main bus in the extreme right enclosure main bus compartment.

3.3 CONTROLLER UNITS

A controller unit is an across-the-line ac magnetic starter with protective devices mounted on a steel framework (saddle) suitable for drawout mounting. Figure 3-1 shows two typical controller units.

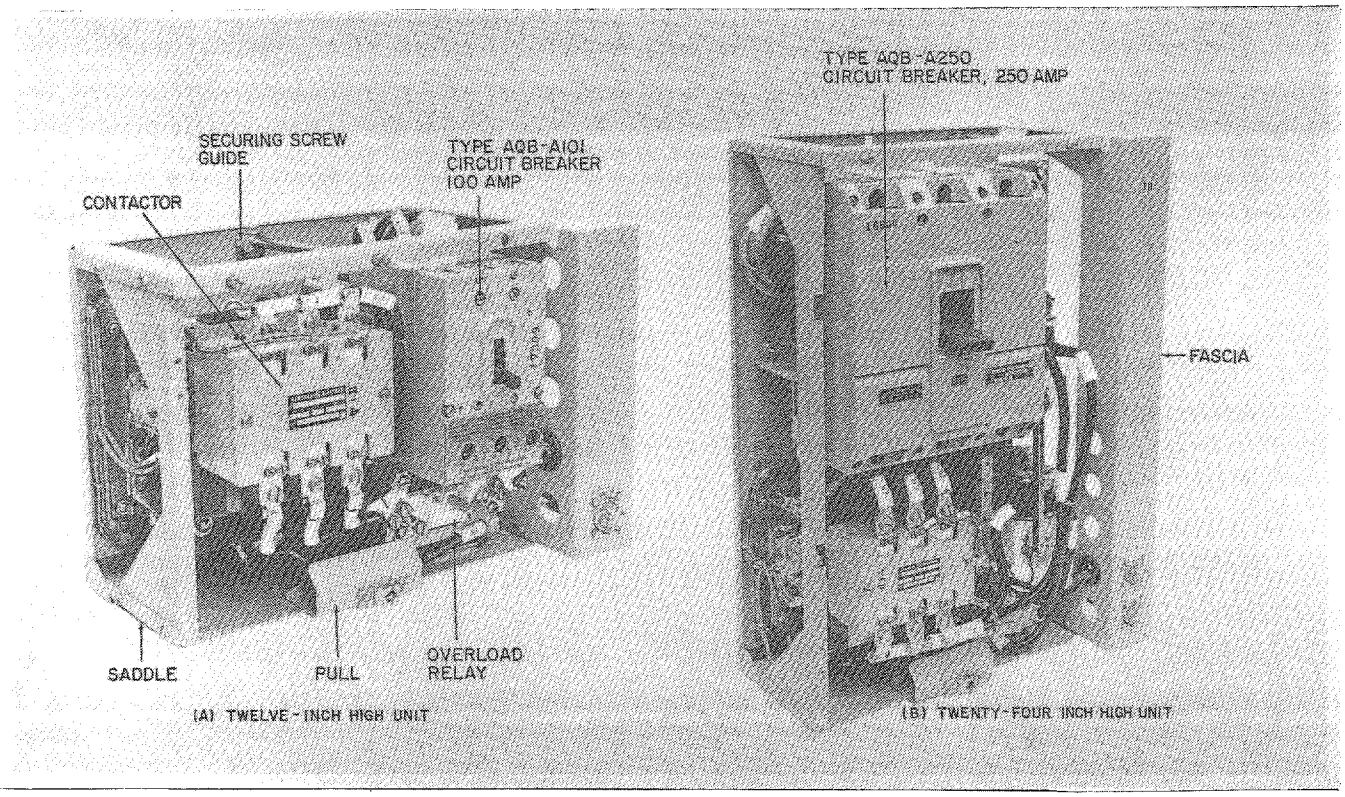


Fig. 3-1. Front view of typical controller unit

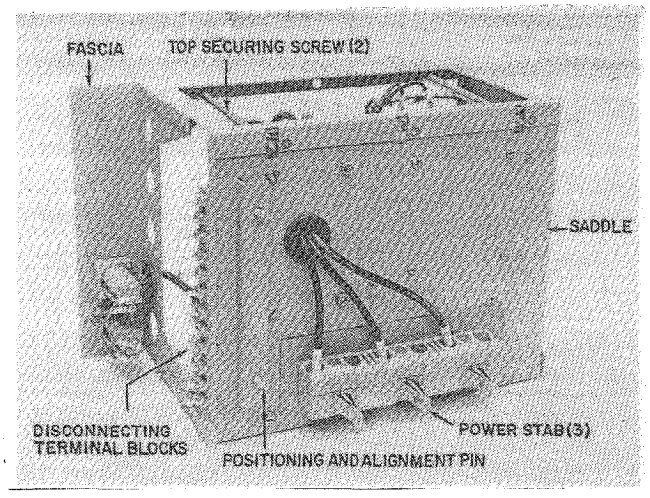


Fig. 3-2. Back view of typical controller unit, twelve inch high

All controller units are fully wired to disconnecting terminal blocks furnished with the unit. The fixed portion of these two-piece terminal blocks is fastened to the enclosure. Simultaneous engagement or disengagement of terminal blocks occurs as the unit is inserted or removed.

The fascia mounts the pilot devices, indicating fuses and indicating lights. The controller units of the same height and design are readily interchangeable. Figure 3-2 illustrates the back view of a typical controller unit. Power stabs fit matched slots inside the vertical bus. These wedge-type double spring-steel stabs provide tight, low-resistance bus connections.

3.4 DISTRIBUTION UNITS

A distribution unit contains either one, two, or three circuit breakers of a specified type and rating in accordance with MIL-C-17361 specifications. Figures 3-3 and 3-4 illustrate a typical 100-ampere distribution unit. These circuit breakers contain externally adjustable instantaneous trip settings on each pole and can be changed to suit field needs. Circuit breaker handles are front accessible through cutouts in the cover. Guards are provided for the handles to reduce the possibility of inadvertent operation. A Shipboard application nameplate appears on the cover over each circuit breaker. Power is cabled directly to the line terminals from the main bus; thus, connections to the vertical bus are not required. Grommeted holes are provided in the shelf below the circuit breakers to facilitate cabling to the load.

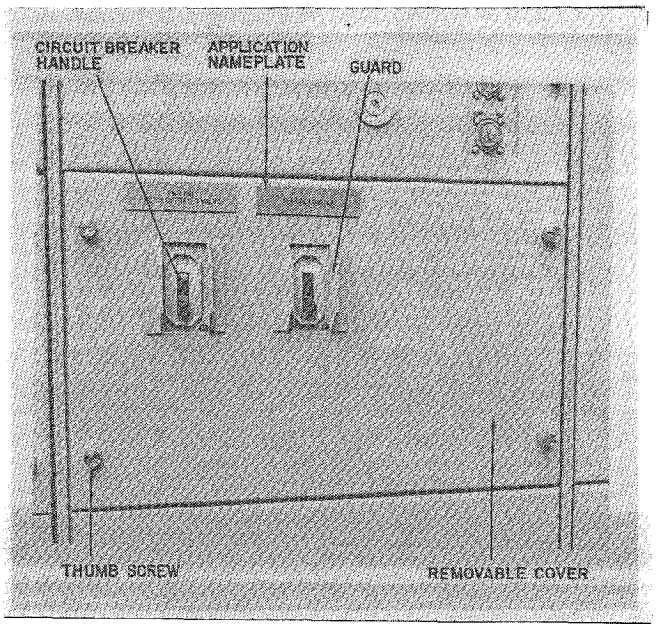


Fig. 3-3. Front view of typical distribution unit

3.5 EQUIPMENT CABLING

Eighty-seven square inches of cable entrance area is available at the bottom of each enclosure for all incoming power and control wiring. Incoming cables terminate either at the individual units (Type B wiring), or at the master terminal boards (Type C wiring), located in the main bus compartment. See paragraph 8.4 for additional cabling instructions.

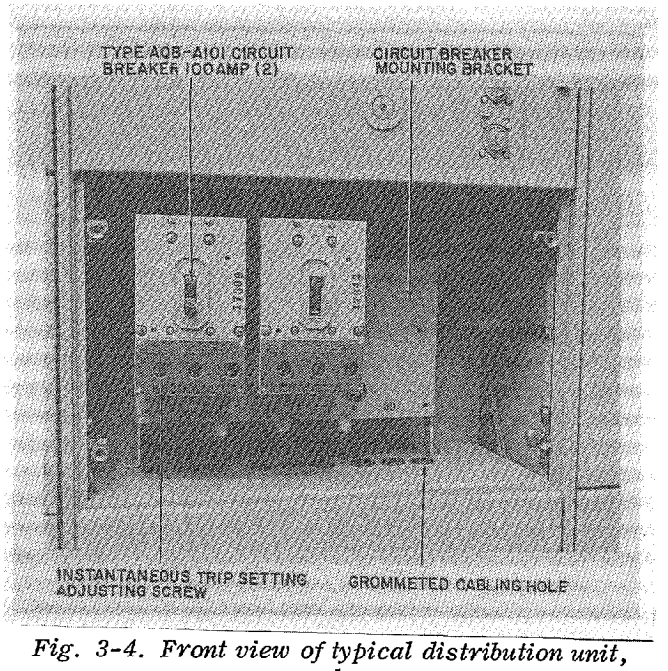


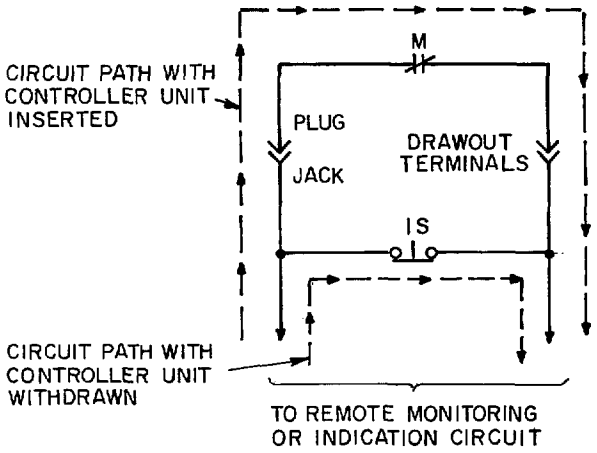
Fig. 3-4. Front view of typical distribution unit, cover removed

3.6 AUXILIARY RELAYS

Auxiliary relays, if required, are located in the main bus compartment and are wired out to terminal boards at the factory.

3.7 DRAWOUT FEATURES

A unique unit guidance system assures engagement of the power stabs to the vertical bus and of the drawout terminal to the fixed terminals. This system consists of a fixed guide (see Fig. 4-2) on the shelf and a mating slot on the controller unit. A shelf latch (see Fig. 4-2) is provided which must first be depressed before the controller unit can be completely withdrawn. A normally closed interlock push button switch (IS) is wired in parallel with each normally closed auxiliary contact wired out through drawout terminals. Function of the interlock switch (IS) is to maintain circuit continuity when the controller unit is withdrawn. This switch is held in the OPEN position with the controller unit fully inserted. Figure 3-5 illustrates the circuit operation.



NOTE:
1. M CONTACT IS NORMALLY CLOSED, AUXILIARY CONTACT IN CONTROLLER UNIT.
2. IS CONTACT IS NORMALLY CLOSED, MOMENTARY CONTACT THAT IS HELD IN OPEN POSITION WITH CONTROLLER UNIT INSERTED.

Fig. 3-5. Interlock switch circuit

SECTION 4
SCHEDULED MAINTENANCE

4.1 CONTROLLER UNITS

Refer to GE Instruction Publication GEK-40589, "Navy AC and DC Magnetic Starters," for scheduled maintenance applicable to controller units.

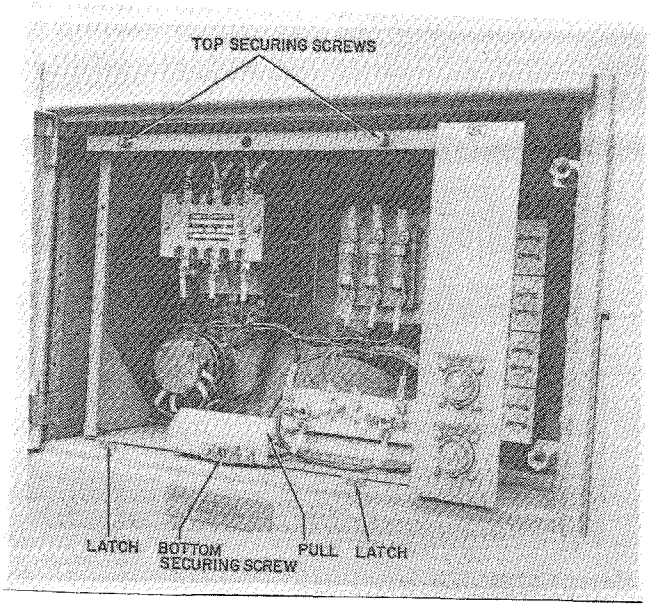


Fig. 4-1. Typical fully inserted controller unit, door open

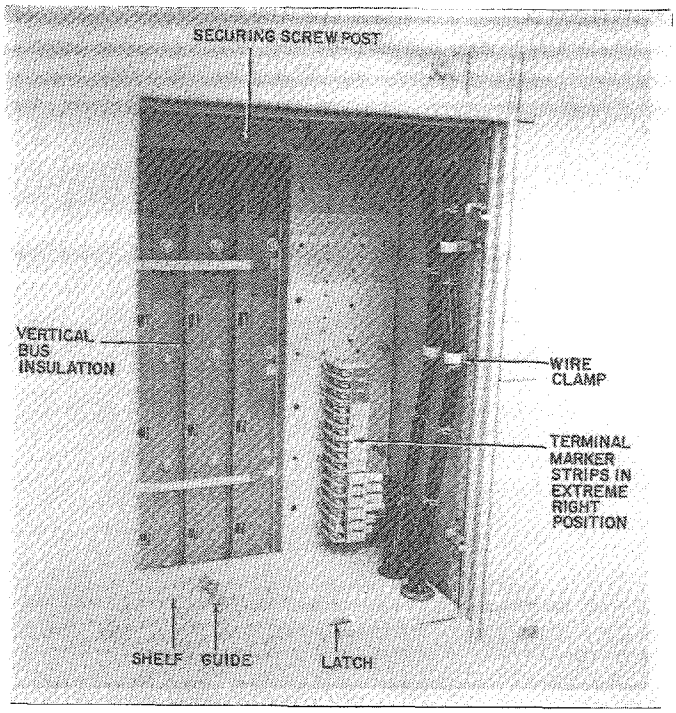


Fig. 4-2. Controller unit compartment, door open

4.2 CIRCUIT BREAKERS

Scheduled maintenance procedures for circuit breakers are contained in the manufacturer's instruction publications. The design and manufacture of Navy circuit breakers is such that, with proper application, internal servicing of the circuit breaker is not normally necessary.

4.3 CONTROLLER UNIT REMOVAL

The procedure for withdrawing a controller unit is as follows:

1. De-energize the controller with the master switch and turn the circuit breaker, if used, to the OFF position.
2. Loosen the door thumb screws and open the door.
3. Unscrew the securing screws located at the top and bottom of the controller unit. The three securing screws are identified in Fig. 4-1.
4. Grasp the pull (see Fig. 4-1) and pull outward; this will disengage all unit terminal stabs from the vertical bus and stationary terminal blocks.
5. Continue to pull the unit outward until further movement is restricted by the latches.
6. The unit is completely withdrawn from the compartment by simultaneously releasing the spring loaded latches and sliding unit forward; take care that the unit does not drop as it is withdrawn.

4.4 CONTROLLER UNIT REPLACEMENT

The procedure for replacing a controller unit is as follows:

1. Loosen the door thumb screws and open the door.
2. The sliding terminal marker strips must be in the extreme right position with all screws tightened. Figure 4-2 shows the correct position of marker strips for inserting unit.
3. Place the unit on shelf. Align guidance slot on unit with guide located on shelf. Push unit inward until bottom securing screw can be started.
4. Turn the bottom securing screw using an 18-inch heavy-duty screwdriver to obtain engagement of unit stabs.
5. Turn the top left securing screw to draw in unit. Next, begin turning the right securing screw. If misalignment prevents either of the top securing screws from engaging its matching post (see Fig. 4-2), adjust the securing screw guide (see Fig. 3-1a) by loosening the set screw, thus enabling the guide to "float." Retighten the set screw after obtaining correct alignment.
6. Alternately continue tightening all three securing screws until the unit is fully racked into the operating position.

SECTION 5
TROUBLESHOOTING

5.1 GENERAL

Troubleshooting techniques for individual magnetic starters are also applicable to control center starters. Refer to GE instruction publication GEK-40589, "Navy AC and DC Magnetic Starters," for troubleshooting assistance.

5.2 TROUBLESHOOTING CHART

Troubleshooting chart, Table 5-1, applies specifically to control centers and supplements troubleshooting procedures in GE instruction publication GEK-40589.

TABLE 5-1
TROUBLESHOOTING CHART

PROBLEM	CAUSE	REMEDY
INTERLOCK SWITCH Failure to Operate	(a) Defective switch. (b) Setscrew improperly adjusted.	(a) Replace switch. (b) Adjust setscrew.
CONTROLLER UNIT Difficult to Rack into Operating Position	(a) Mating terminal stabs misaligned. (b) Top securing screw(s) misaligned with matching post(s). (c) Top securing screw(s) doesn't reach matching post(s).	(a) Align fixed terminal stabs. (b) Adjust securing screw(s). (c) Further tighten bottom securing screw and exert inward force upon top left corner of unit.
OVERLOAD RELAYS Failure to Trip	Unit not fully inserted into operating position. Relay remains in reset position (rod depressed).	Tighten unit securing screws.
DOORS Sticking or Binding	Door hinge out of adjustment.	Loosen hinges, re-position door, tighten hinges.

SECTION 6
CORRECTIVE MAINTENANCE

6.1 INTRODUCTION

The objective of this section is to refer Navy maintenance personnel to the appropriate publications which contain instructions required to perform corrective maintenance tasks.

6.2 CONTROLLER UNIT

General Electric instruction publication GEK-40589, "Navy AC and DC Magnetic Starters," contains information on adjustments, repairs, and replacement.

6.3 INDIVIDUAL DEVICE REPAIR

Detailed individual device repair procedures are contained in the appropriate device instruction publications.

6.4 CIRCUIT BREAKER MAINTENANCE

See manufacturer's instruction publications for corrective maintenance that may be accomplished in the field.

SECTION 7
PARTS LIST

7.1 GENERAL

A complete tabulation of repair parts for General Electric devices is contained in GE instruction publication GEK-40591, "Parts List for Navy AC and DC Magnetic Starters." Repair parts information covering circuit breakers is contained in the appropriate manufacturer's instruction publication.

SECTION 8
INSTALLATION

8.1 GENERAL

An inspection should be made for damage of the equipment in transit and corrective action should be taken as soon as possible after receipt of the equipment.

8.2 SHIPPING AND HANDLING

The equipment is shipped with two 1 1/2-inch by 3-inch steel channel floor sills and a 2-inch by 3-inch lifting angle (see Fig. 8-1). The equipment is shipped in an upright position and this position should be maintained during handling. Never attempt to jack, lift, or move the equipment at points other than the lifting angle or floor sills. The floor sills and lifting angle should only be removed upon installation.

CAUTION: *BEFORE MOVING EQUIPMENT, BE SURE THE DOORS ARE CLOSED AND THEIR THUMB SCREWS FASTENED.*

8.2.1 Floor Sills

The floor sills are removed by opening the doors to the main bus compartment and removing the bolts which secure the floor sills to the bottom of the equipment.

8.2.2 Lifting Angle

The lifting angle is shown in Fig. 8-1 and is bolted through 1-inch thick washers into weld nuts (No. 1/2-13, two per vertical section) on the inside top of the equipment. Lifting holes are 1 1/2-inch diameter. After removing the lifting angle, the bolts and other mounting hardware should be retained for use in attaching sway bracing.

8.3 MOUNTING

Six 25/32-inch diameter holes per each vertical section are provided for bolting the equipment to the deck. Consult the outline and floor plan drawings furnished by the General Electric Company for the mounting details. Sway bracing should be attached on the top of the equipment using the tapped holes provided for the lifting angle (see paragraph 8.2.2 for access to these holes).

It is not advisable to make mounting provisions based upon drawing dimensions. Alignment difficulties may be experienced due to normal factory tolerances of ± 1/16-inch on mounting dimensions and cummulative tolerance buildup on multiple section lineups.

8.4 INCOMING CABLING

Main incoming power cables should be brought through the bottom of the extreme right vertical section and connected to the main bus. The main bus contains two slotted holes per phase for through bolting of the main incoming power cable lugs to the main bus. Figure 8-2 shows these provisions.

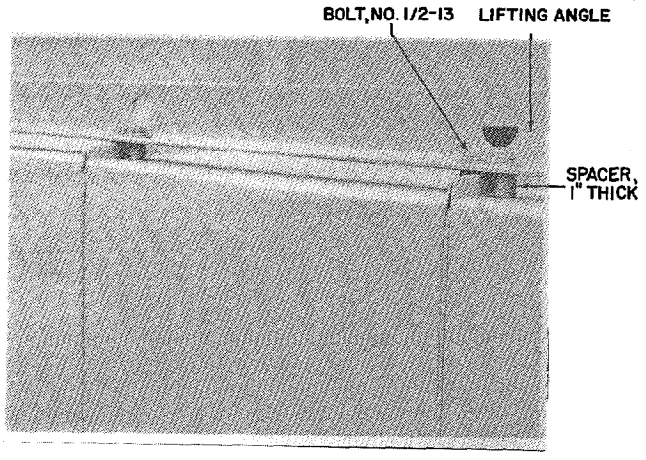


Fig. 8-1. Rear view of control center showing lifting angle

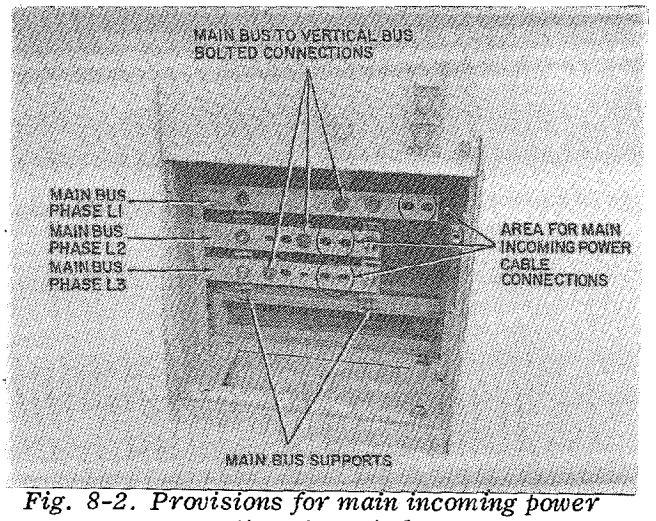


Fig. 8-2. Provisions for main incoming power connections in main bus compartment, door open

Whenever master terminal boards (Type C wiring) are provided, main incoming power cable connections should be made first before the installation of any other incoming wiring. This is important because it is necessary to swing the master terminal boards, terminal-board mounting straps, and factory-installed wiring outward (away from the main bus) as an integral assembly, for maximum accessibility to the main bus. The main bus is located immediately in back of the master terminal boards. This procedure requires the removal and replacement of all terminal-board mounting-strap installation hardware.

Terminals are provided for all other power and control incoming ship's cables either at master terminal boards, located in the main bus compartment (Type C wiring), or at terminals located near drawout units (Type B wiring). Consult equipment drawings furnished by the General Electric Company for actual location of terminal boards. Terminals are numbered on the equipment as indicated on the drawings and door diagrams.

Wiring to individual units should be fed through knockouts provided in shelves. Small diameter wires may be fed through the space between the edge of shelf and the right side of enclosure. Wiring should be neatly bundled, wrapped, and securely clamped to brackets mounted along the right-hand side of the section. Figure 4-2 illustrates the proper method of installing vertical wiring.

Attention should be given, when installing cables, to avoid the possibility of cable insulation being damaged by any sharp edges (steelwork, screws, etc.).

8.5 DOOR AND COVER DIAGRAMS

Plastic laminated diagrams, showing the circuit schematics, connections, and other pertinent data, are secured to the inside of the doors and covers.

8.6 PREPARATION FOR INITIAL OPERATION

See GE instruction publication GEK-40589, "Navy AC and DC Magnetic Starters," paragraph 8.4, for preparatory check list.

GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS OPERATION
SALEM, VA. 24153
FSMC 02989

GENERAL ELECTRIC

PRINTED IN U.S.A.